

Regularized Multivariate Two-way Functional Principal Component Analysis

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Outline

1. **Introduction & Background**
2. **A SVD Approach for Regularized Multivariate FPCA**
3. **Two-way Regularized Multivariate FPCA**
4. **Conclusion & Future Work**

Introduction

Functional Data Analysis:

Functional data are observations that vary continuously over a domain (like time, space, or wavelength) and are often visualized as curves, trajectories, or functions rather than isolated points.

These data are often recorded at discrete time points or grid locations, even though they originate from continuous processes in areas like engineering, finance, environmental science, and healthcare.

Functional Data Analysis (FDA) is a statistical framework that treats these observations as realizations of smooth underlying functions, allowing for more accurate modeling and interpretation of continuous processes.

Regularized MFPCA

Two-way Regularized MFPCA

Conclusion & Future Work